Solutions and teaching notes, chapter 3

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $(3; -4)$. *Action:* Subtract start from end on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish between norm and coordinates.
3. $\overrightarrow{AC}$. *Action:* Draw the combined displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross term.
5. $\frac{3}{5}$. *Action:* Name the sides with respect to the angle.
6. $(1; 2)$. *Action:* Average the two coordinates separately.

Investigation

Take points on the circle with diameter AB; proof using the midpoint I: product $MI^2 - 9$. The endpoints are included because one of the vectors is zero.

Direct applications

1. 0 then -10. The cosine is respectively 0 and -1.
2. $3t - 12 = 0$, therefore $t = 4$.
3. The square equals $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, therefore $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: circle with centre (3; 0) and radius 4.

Exercises: methods and answers

Exercise 3.1 - Direct calculations. Product $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum (5; 1) with square 26. Check: $5 + 2(-2) + 25 = 26$.

Exercise 3.2 - A right-angled triangle? $\overrightarrow{AB} = (3; 1)$, $\overrightarrow{AC} = (-1; 3)$; product 0. $AB = AC = \sqrt{10}$, area $\frac{AB \times AC}{2} = 5$ units of area.

Exercise 3.3 - Length without coordinates. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Product $\frac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$.

Exercise 3.4 - Finding an angle. $\cos\theta = -\frac{6}{12} = -\frac{1}{2}$; in [0; 180 °], the angle is 120°. The domain ensures uniqueness.

Exercise 3.5 - A variable locus. $MI^2 = k + \dfrac{AB^2}{4}$. If the right-hand side is negative: the set empty; zero: the single point I; positive: circle whose radius is its square root. The discussion concerns a squared distance.

Exercise 3.6 - Work done by a force. $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. À $120°$, $W = -100$ J. The sign distinguishes a tailwind from a headwind; it does not describe a negative distance.

Exercise 3.7 - Contrapositive. The direct statement "orthogonal implies equality of squares" follows from the expansion. Its contrapositive is exactly the statement. One can also write 2 $\vec{u} \cdot \vec{v} \neq 0$.

Exercise 3.8 - Delayed review. Al-Kashi's theorem: $\dfrac{9 + 16 - 25}{2}$ = 0. Converse of Pythagoras' theorem: triangle right-angled at A, therefore zero product. Name the assumptions of each method.

CQFD, exercise 9 p. 250, questions a and b. a : $1 \times 3 + 2 \times (-1) = 1$. b : $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. $\overrightarrow{AB} = (4; 6)$, $\overrightarrow{AC} = (-3; 4)$, $I(-1; -2)$ and $\overrightarrow{CI} = (5; -1)$. Hence $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI}$ = 14. Chasles' relation gives $\overrightarrow{AB} \cdot \overrightarrow{AI}$ = 12 + 14 = 26. Two checks: $\overrightarrow{AI}$ = (2; 3) gives 8 + 18 = 26; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16 + 36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.

2. C. Square root of 25.

3. D. Scalar criterion.

4. B. $\dfrac{2 \times 3}{2} = 3$.

5. C. Expand the scalar square.

MCQ 2.

1. D. Negative cosine.

2. A. Coordinates by projection.

3. C. $MI^2 = \dfrac{AB^2}{4}$.

4. B. Bilinearity: 2(-1)3.

5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.

2. True: its product with any vector is 0.

3. False: (1; 0) and (0; 1) are non-zero and orthogonal.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.